When Not to Trust a Traffic Forecast: Shift- and Outage-Aware Risk-Controlled Spatiotemporal Learning for Intelligent Transportation Systems

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Abstract

Traffic forecasting benchmarks primarily reward average point accuracy under a single chronological test split, whereas deployed intelligent transportation systems face compound failures: traffic regimes change while spatially and temporally correlated sensors become unavailable. We study forecasting as an operational decision problem in which every forecast origin requires a point tensor, node–horizon calibrated intervals, and an option to abstain. We introduce ShiftGuard, a reliability-conditioned graph–temporal forecaster with structured-outage augmentation, heterogeneity-aware scale estimation, and prequential group-adaptive conformal risk control. We also define a leakage-controlled compound-shift protocol for METR-LA and PEMS-BAY covering independent, burst, contiguous-region, and hub outages across traffic regimes. The evaluation comprises 640 confirmatory stress runs in addition to clean tests, using five seeds. Averaged over four non-clean outage families and five severities, ShiftGuard reduces MAE relative to the strongest local transformer from 3.74 to 3.40 mph on METR-LA and from 2.49 to 1.94 mph on PEMS-BAY; crossed seed–scenario bootstrap intervals for the differences are $[-0.465, -0.234]$ and $[-0.682, -0.411]$ mph. At 30% outage severity, risk control retains 86% and 98% of forecasts while limiting empirical origin-level failure to 8% and 2%, respectively. Clean PEMS-BAY MAE is 0.027 mph worse than STAEformer, so the contribution is robustness and actionable uncertainty rather than universal clean-error dominance.

Keywords: traffic forecasting, intelligent transportation systems, spatiotemporal graph neural network, sensor outage, distribution shift, conformal prediction, selective prediction

1. Introduction

Short-term traffic forecasts support routing, ramp metering, incident response, and traveler information. Graph neural forecasters have improved canonical accuracy by modeling diffusion [1], learned dependencies [2], node heterogeneity [3], and dynamic meta-graphs [4]. More recent results show that simple identity embeddings [5] and vanilla transformers with adaptive embeddings [6] can rival elaborate designs. This progress, however, is measured mainly by average MAE, RMSE, and MAPE on clean chronological test blocks.

Average error is not a sufficient safety signal. A model can be accurate in free flow yet fail during congestion transitions; marginal coverage can conceal undercoverage at particular sensors or horizons; and a calibrated clean-data interval can become unreliable when an upstream sensor region disappears. The operational question is therefore not only "what is the future speed?" but also "how uncertain is this forecast, and should an automated consumer act on it?"

This work makes four contributions:

1. a reproducible compound-shift protocol that crosses traffic regimes with four graph-aware sensor-outage families;
2. a compact graph–temporal backbone trained with graph-structured outage augmentation and explicit observation masks;
3. node–horizon adaptive conformal calibration and a selective decision rule evaluated through worst-group coverage and risk–coverage; and
4. an accuracy–robustness–reliability–efficiency comparison that keeps locally reproduced and literature-reported results strictly separate.

2. Related work

2.1. Deterministic traffic forecasting

DCRNN [1] models directed diffusion with recurrent sequence learning, while STGCN [7] and Graph WaveNet [2] combine graph operations with temporal convolutions. AGCRN [3] learns node-adaptive parameters. MegaCRN [4] addresses spatiotemporal heterogeneity through a meta-node bank. PDFormer [8] captures propagation delay and long-range dependence. STID [5] and STAEformer [6] demonstrate that careful identity and adaptive embeddings are unusually strong, motivating simple baselines

and explicit cost reporting. A recent benchmark audit further shows that spatially unaware linear models can be stronger than commonly reported, so we include persistence and autoregressive ridge controls under the same split and preprocessing [9]. GMAN uses spatiotemporal attention [10]; MTGNN and GTS learn latent graph structure [11, 12]; and ST-Norm separates spatial and temporal normalization [13]. These lines of work make another graph block, by itself, an insufficient novelty claim. TESTAM further routes recurring and non-recurring patterns among specialized experts [14], motivating the regime router ablation in this study.

2.2. Nonstationarity, missingness, and uncertainty

Test-time compensation and dynamic graph learning target extremes or nonstationarity; CompFormer retrieves historical representations for extreme traffic [15]. CAST-GNN instead targets weekday-to-unseen- context generalization through causal invariance and dynamic associations [16]. A concurrent regime audit reports that aggregate accuracy and marginal coverage can conceal severe transition-regime failures on METR-LA and PEMS-BAY [17]. Masked pretraining improves robustness and context learning through spatial and temporal reconstruction [18, 19], while imputation-oriented models address incomplete traffic signals. These problems are frequently evaluated separately. SC-SAMG studies held-out sensors, short support-set adaptation, and outage robustness, but its problem is cold-start point forecasting rather than selective reliability under simultaneous temporal and measurement shifts [20]. A systematic spatiotemporal uncertainty study compares Bayesian and frequentist routes [21]. Conformal time-series forecasting provides multi-horizon intervals under explicit assumptions [22]. Online conformal methods address changing distributions [23], while CAP studies inference after adaptive selection [24]. Patil et al. combine availability-aware graph construction with adaptive conformal intervals for traffic forecasting [25]; however, they do not evaluate graph-correlated outage families, worst-group reliability, or controlled abstention. Most recently, CSTH conditions conformal interval scale on hypergraph density and volatility, with blocked-time calibration on METR-LA and PEMS-BAY [26]. It does not study structured sensor outages, delayed prequential recalibration, or selective-risk control. Marginal coverage is nevertheless not equivalent to reliable node-, horizon-, and regime-level decisions. Our focus is their compound interaction and the risk–coverage trade-off produced when a traffic forecaster may abstain.

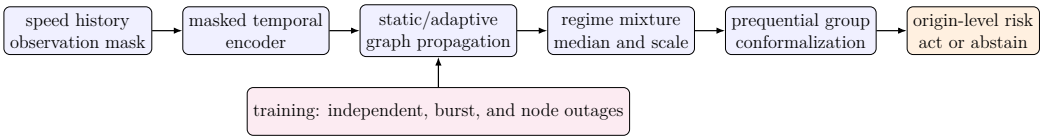


Figure 1: ShiftGuard training and deployment pipeline. Calibration and risk decisions use only validation data and previously resolved forecast origins.

3. Problem formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$ contain N sensors. At time t, $X_t \in \mathbb{R}^{P \times N}$ contains the previous $P = 12$ speed observations and $M_t \in \{0,1\}^{P \times N}$ their availability. For $H = 12$, the system returns point forecasts $\hat{Y}_t \in \mathbb{R}^{H \times N}$, intervals $[L_t, U_t]$, and a forecast-origin decision $D_t \in \{0,1\}$. We seek low point risk and narrow calibrated intervals while controlling selective risk under an environment e_t and outage process M_t that can differ from training.

4. Method

4.1. Reliability-conditioned encoding

Each node concatenates standardized speed and its binary observation mask over the input window. A shared temporal encoder, learned node identity, and a direct history-to-horizon residual path produce z_i and preserve a strong simple forecasting route. Node reliability $r_i = P^{-1} \sum_p M_{p,i}$ modulates outgoing graph messages. Static and learned adaptive adjacency are mixed, sender-weighted by r_i, and row normalized. Gated residual graph blocks prevent unreliable messages from dominating local representations.

4.2. Regime routing and probabilistic head

Global level, volatility, absolute first difference, and availability summarize the current context. A small router mixes K output experts. Each expert emits a median residual relative to persistence and a positive Laplace scale. Training uses Laplace negative log likelihood under a randomized curriculum of independent, burst, and node outages.

4.3. Conformal risk control

Calibration scores normalize absolute errors by the predicted scale. Quantiles are estimated separately for each node and horizon using only prior

Table 1: Dataset diagnostics computed from checksum-verified raw files. Speed is in miles per hour; correlations use observed values at 5-minute and one-day lags.

Dataset	Timestamps	Sensors	Missing (pct.)	Mean speed	Std. speed	Lag-1 corr.	Lag-288 corr.
METR-LA	34272	207	8.109	58.460	13.024	0.942	0.627
PEMS-BAY	52116	325	0.003	62.621	9.588	0.983	0.681

residuals, with shrinkage toward a global quantile for data-poor groups. Interval width is the primary abstention score. Sorting forecasts by this score yields a risk–coverage curve and area under the curve (AURC). For an operational error tolerance, the accepted set is chosen as the largest candidate coverage whose Bonferroni-corrected exact binomial upper bound is below a registered risk limit. The operational unit is a forecast origin: its error and uncertainty are respectively the mean absolute error and mean interval width over all node–horizon cells. Accept/reject decisions therefore apply to whole forecast tensors rather than treating correlated cells as independent trials. Candidate thresholds are constructed from even past forecast origins and certified on disjoint odd past origins; current-block targets never participate. The simultaneous bound is exact when the certified accepted origin-level failure indicators satisfy the stated i.i.d. Bernoulli assumption. The paper distinguishes empirical prequential coverage under drift from finite-sample exchangeability guarantees; no guarantee is claimed when assumptions are violated.

5. Experimental design

5.1. Datasets and leakage control

METR-LA contains 207 sensors and PEMS-BAY 325 sensors at five-minute resolution. Raw time is split chronologically 70/10/20. Normalization is fitted on the training block only. Twelve historical steps predict the next twelve. Validation and test windows may use preceding context, but targets never cross backward across a split boundary.

5.2. Baselines

Locally reproduced baselines include historical average, persistence, ridge AR, linear, MLP/GRU, STGCN, DCRNN, Graph WaveNet, AGCRN, STID, and a faithful STAEformer configuration. Compatible open implementations of recent methods are added after preprocessing and license audits. Reported

Table 2: Literature-reported point accuracy for context only. Protocol and implementation differences prohibit direct significance comparisons with locally reproduced results.

Dataset	Model	MAE at 15 min	MAE at 30 min	MAE at 60 min	Average MAE
METR-LA	DCRNN	2.77	3.15	3.60	–
METR-LA	Graph WaveNet	2.69	3.08	3.51	–
METR-LA	STAEformer	2.65	2.97	3.34	–
METR-LA	TESTAM	–	–	–	2.93
PEMS-BAY	DCRNN	1.38	1.74	2.07	–
PEMS-BAY	Graph WaveNet	1.30	1.63	1.99	–
PEMS-BAY	STAEformer	1.31	1.62	1.88	–
PEMS-BAY	TESTAM	–	–	–	1.53

SOTA values appear in a separate contextual table and are never used for significance claims. The recurrent and graph baselines are protocol-matched compact reproductions, not claims of bitwise equivalence to their authors' code; a model-by-model deviation audit accompanies the repository.

5.3. Stress tests and metrics

We cross clean, independent, temporal-burst, contiguous-region, and high-degree hub outages at 10–50% severity with free-flow, congested, stable intermediate, and transition regimes. Metrics include horizon-specific and average MAE, RMSE, MAPE; degradation and worst-regime MAE; coverage, width, Winkler score and worst-group coverage; AURC and selective MAE; parameters, FLOPs, latency, peak memory, and training time. Headline results use at least five seeds and paired bootstrap confidence intervals.

The confirmatory environment is captured automatically, including Python and package versions, CUDA build, GPU, and driver. Runs use PyTorch 2.13.0 with CUDA 13.0 on an NVIDIA GeForce RTX 5060 Laptop GPU; the generated environment manifest is retained with the result artifacts.

6. Results

6.1. Canonical accuracy

Table 3 is generated exclusively from local experiment artifacts. A stochastic model enters the confirmatory table only after all five registered seeds complete; partial runs are omitted.

Table 3: Locally reproduced canonical-test results. Empty uncertainty cells indicate that only one deterministic run is applicable or currently complete.

Dataset	Model	MAE	SD	RMSE	MAPE (pct.)	Parameters	Train (s)	Seeds
METR-LA	STAEformer	3.032	0.008	6.178	8.362	1258932	4554.221	5
METR-LA	ShiftGuard	3.033	0.005	6.087	8.511	89012	357.521	5
METR-LA	DCRNN	3.247	0.010	6.380	9.133	63372	1002.598	5
METR-LA	STID	3.273	0.015	6.603	9.558	41996	90.004	5
METR-LA	Graph WaveNet	3.285	0.008	6.451	9.205	22744	749.354	5
METR-LA	AGCRN	3.343	0.010	6.576	9.494	16764	492.095	5
METR-LA	GRU	3.761	0.002	7.628	10.878	13644	464.916	5
METR-LA	MLP	3.798	0.002	7.662	11.032	5772	68.023	5
METR-LA	STGCN	3.899	0.013	7.187	11.365	17548	374.899	5
METR-LA	Linear	3.982	0.001	7.728	11.589	156	48.049	5
METR-LA	Historical average	4.189	–	7.856	13.044	0	0.000	1
METR-LA	Persistence	4.244	–	8.154	11.052	0	0.000	1
METR-LA	Ridge AR	4.294	–	7.595	12.727	324	0.093	1
PEMS-BAY	STAEformer	1.577	0.006	3.650	3.613	1372212	13896.209	5
PEMS-BAY	ShiftGuard	1.604	0.003	3.591	3.691	100340	860.822	5
PEMS-BAY	STID	1.664	0.004	3.739	3.822	49548	199.245	5
PEMS-BAY	DCRNN	1.683	0.005	3.779	3.900	63372	2888.655	5
PEMS-BAY	Graph WaveNet	1.720	0.005	3.770	3.934	25104	1830.569	5
PEMS-BAY	AGCRN	1.731	0.004	3.849	4.016	18652	1338.461	5
PEMS-BAY	GRU	1.955	0.001	4.619	4.566	13644	1081.630	5
PEMS-BAY	MLP	1.969	0.004	4.648	4.569	5772	169.266	5
PEMS-BAY	Linear	2.130	0.001	4.955	4.808	156	71.894	5
PEMS-BAY	Persistence	2.171	–	5.125	4.671	0	0.000	1
PEMS-BAY	Ridge AR	2.302	–	4.831	5.371	324	0.086	1
PEMS-BAY	STGCN	2.506	0.005	4.873	5.918	17548	903.628	5
PEMS-BAY	Historical average	2.630	–	5.228	6.176	0	0.000	1

Paired inference uses non-overlapping daily blocks of forecast origins. The reported confidence interval is a crossed seed–day paired-bootstrap interval for $\Delta\text{MAE} = \text{MAE}_{\text{ShiftGuard}} - \text{MAE}_{\text{comparator}}$; both seeds and chronological daily blocks are resampled. Wilcoxon tests operate on paired daily-block differences averaged over seeds and are Holm corrected across the displayed comparisons.

For METR-LA, the completed five-seed comparison does not establish clean-error superiority over the local STAEformer implementation: the origin-weighted MAE difference is -0.0007 mph (crossed-bootstrap 95% interval $[-0.0308, 0.0221]$ mph; Holm-adjusted $p = 0.944$). This statistical estimand weights forecast origins equally, whereas the canonical table pools observed node–horizon errors. Failure to detect a difference is not proof of equivalence. The proposed reliability contribution therefore requires independent support from the registered outage and selective-prediction

Table 4: Paired daily-block statistical comparison. Negative ΔMAE favors ShiftGuard.

Dataset	Comparator	Delta MAE	CI low	CI high	Improvement (pct.)	Holm p
METR-LA	agcrn	-0.3160	-0.3607	-0.2801	9.3884	< 0.0001
METR-LA	dcrnn	-0.2188	-0.2621	-0.1977	6.6928	< 0.0001
METR-LA	graphwavenet	-0.2559	-0.2867	-0.2267	7.7398	< 0.0001
METR-LA	gru	-0.7292	-0.8130	-0.6167	19.2941	< 0.0001
METR-LA	historical average	-1.1510	-1.3156	-1.0603	27.3965	< 0.0001
METR-LA	linear	-0.9513	-1.0477	-0.8318	23.7737	< 0.0001
METR-LA	mlp	-0.7707	-0.8650	-0.6721	20.1695	< 0.0001
METR-LA	persistence	-1.2139	-1.2951	-1.1067	28.4667	< 0.0001
METR-LA	ridge ar	-1.2623	-1.3503	-1.1482	29.2695	< 0.0001
METR-LA	staeformer	-0.0007	-0.0314	0.0216	0.0228	0.9441
METR-LA	stgcn	-0.8670	-0.9338	-0.8072	22.1324	< 0.0001
METR-LA	stid	-0.2431	-0.3038	-0.2065	7.3817	< 0.0001
PEMS-BAY	agcrn	-0.1264	-0.1455	-0.1036	7.3042	< 0.0001
PEMS-BAY	dcrnn	-0.0785	-0.0943	-0.0600	4.6670	< 0.0001
PEMS-BAY	graphwavenet	-0.1155	-0.1301	-0.0972	6.7159	< 0.0001
PEMS-BAY	gru	-0.3509	-0.4114	-0.2761	17.9473	< 0.0001
PEMS-BAY	historical average	-1.0262	-1.1972	-0.8952	39.0150	< 0.0001
PEMS-BAY	linear	-0.5258	-0.6103	-0.4193	24.6876	< 0.0001
PEMS-BAY	mlp	-0.3650	-0.4277	-0.2857	18.5373	< 0.0001
PEMS-BAY	persistence	-0.5664	-0.6506	-0.4624	26.0971	< 0.0001
PEMS-BAY	ridge ar	-0.6982	-0.7884	-0.5839	30.3263	< 0.0001
PEMS-BAY	staeformer	0.0269	0.0158	0.0376	-1.7064	< 0.0001
PEMS-BAY	stgcn	-0.9017	-0.9601	-0.8289	35.9857	< 0.0001
PEMS-BAY	stid	-0.0599	-0.0868	-0.0400	3.6001	< 0.0001

experiments. On PEMS-BAY, STAEformer is better on the clean test: the origin-weighted difference is 0.0269 mph, with a 95% interval $[0.0155, 0.0374]$ and Holm-adjusted $p = 4.93 \times 10^{-6}$. Thus ShiftGuard is not presented as a new clean-data point-forecasting SOTA.

6.2. Compound-shift robustness and calibration

Table 5 averages the registered 30% non-clean outage families; all family-by-rate results remain available in machine-readable artifacts.

Across independent, burst, region, and hub outages at 10, 20, 30, 40, and 50%, the mean ShiftGuard–STAEformer MAE difference is -0.347 mph on METR-LA (crossed seed–scenario 95% interval $[-0.465, -0.234]$) and -0.546 mph on PEMS-BAY ($[-0.682, -0.411]$). Corresponding AURC differences are -1.46 and -1.26. At 30%, worst node–horizon coverage is 0.831 versus 0.816 on METR-LA and 0.851 versus 0.834 on PEMS-BAY; worst-regime coverage is 0.882 versus 0.809 and 0.887 versus 0.701. Seed-level Wilcoxon tests cannot reject after ten-comparison Holm correction ($p_{\text{Holm}} = 0.625$),

Table 5: Compound-shift point, interval, and selective reliability. PI is the 90% prediction interval; failure denotes accepted forecasts exceeding the registered 5 mph error tolerance. Worst node–horizon (N–H) and regime coverage are reported for ShiftGuard and its strongest local comparator.

Dataset	Model	MAE	PI coverage	Worst N-H cov.	Worst regime cov.	AURC	Retained	Failure rate	Seeds
METR-LA	AGCRN	4.729	0.891	–	–	4.950	0.000	–	5
METR-LA	DCRNN	4.629	0.891	–	–	4.791	0.000	–	5
METR-LA	Graph WaveNet	4.895	0.890	–	–	5.125	0.000	–	5
METR-LA	GRU	4.919	0.893	–	–	5.181	0.000	–	5
METR-LA	Linear	5.109	0.894	–	–	5.356	0.000	–	5
METR-LA	MLP	5.054	0.893	–	–	5.301	0.000	–	5
METR-LA	Persistence	5.478	0.895	–	–	5.703	0.000	–	5
METR-LA	ShiftGuard	3.399	0.898	0.831	0.882	2.400	0.864	0.078	5
METR-LA	STAEformer	3.760	0.894	0.816	0.809	3.873	0.055	0.235	5
METR-LA	STGCN	5.309	0.893	–	–	5.536	0.000	–	5
METR-LA	STID	4.092	0.894	–	–	4.256	0.000	–	5
PEMS-BAY	AGCRN	3.073	0.890	–	–	2.952	0.174	0.115	5
PEMS-BAY	DCRNN	2.956	0.889	–	–	2.844	0.183	0.109	5
PEMS-BAY	Graph WaveNet	2.928	0.889	–	–	2.824	0.203	0.102	5
PEMS-BAY	GRU	3.109	0.893	–	–	3.074	0.167	0.126	5
PEMS-BAY	Linear	3.199	0.894	–	–	3.221	0.077	0.203	5
PEMS-BAY	MLP	3.135	0.893	–	–	3.105	0.167	0.127	5
PEMS-BAY	Persistence	3.283	0.894	–	–	3.268	0.001	0.267	5
PEMS-BAY	ShiftGuard	1.948	0.898	0.851	0.887	1.130	0.978	0.020	5
PEMS-BAY	STAEformer	2.516	0.890	0.834	0.701	2.402	0.442	0.085	5
PEMS-BAY	STGCN	3.857	0.893	–	–	3.752	0.073	0.221	5
PEMS-BAY	STID	2.649	0.891	–	–	2.544	0.209	0.099	5

whose attainable resolution is limited by five seeds; we therefore report the hierarchical intervals and effects but do not claim adjusted rank-test significance.

6.3. Selective forecasting, ablation, and efficiency

At 30% severity, prequential risk control reduces accepted-origin failure on METR-LA from 15% without abstention to 8%, while retaining 86% of forecasts. On PEMS-BAY it reduces failure from 4% to 2% at 98% retention. Static calibration fails under drift on METR-LA (15% failure despite full retention), supporting online updates. Table 6 reports all registered calibration controls.

Removing structured augmentation increases MAE by 0.306 mph on METR-LA and 0.703 mph on PEMS-BAY; independent masking alone also loses 0.098 and 0.220 mph. Conversely, removing reliability-conditioned propagation improves MAE by 0.143 and 0.067 mph, and removing the router has a small favorable effect. Table 7 therefore supports structured augmentation as the effective architectural contribution but falsifies the stronger claim that every proposed mechanism is necessary.

ShiftGuard uses 89k/100k parameters on METR-LA/PEMS-BAY versus

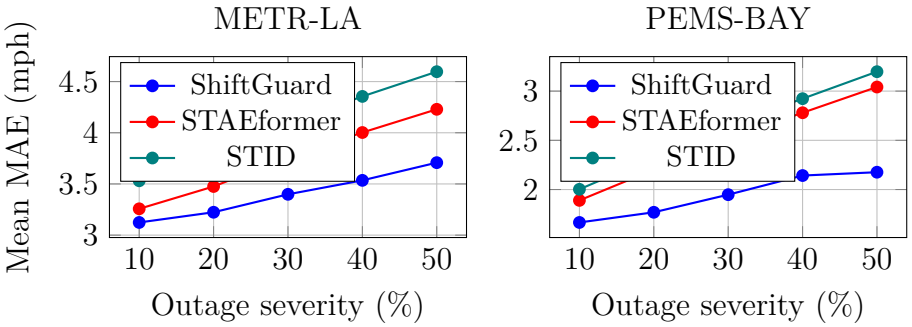


Figure 2: MAE averaged over independent, burst, region, and hub outages and five seeds. ShiftGuard's advantage increases with outage severity.

Table 6: Calibration and selection controls at 30% outage severity, averaged over non-clean families and five seeds.

Dataset	Calibration	Selection	PI coverage	Retained	Failure rate
METR-LA	global-prequential	risk	0.898	0.864	0.079
METR-LA	group-prequential	all	0.898	1.000	0.154
METR-LA	group-prequential	risk	0.898	0.864	0.078
METR-LA	group-static	risk	0.893	0.996	0.152
PEMS-BAY	global-prequential	risk	0.898	0.977	0.020
PEMS-BAY	group-prequential	all	0.898	1.000	0.037
PEMS-BAY	group-prequential	risk	0.898	0.978	0.020
PEMS-BAY	group-static	risk	0.884	0.950	0.015

1.26M/1.37M for STAEformer. Its batch-one latency is 2.30/2.43 ms, and its batch-eight throughput is 3567/3335 samples s^{-1}, compared with 503/269 for STAEformer on the same RTX 5060 Laptop GPU. Table 8 reports the principal local efficiency controls.

Table 7: Ablation results at 30% severity, averaged over non-clean families.

Dataset	Variant	MAE	AURC	Retained
METR-LA	ShiftGuard	3.399	2.400	0.864
METR-LA	Adaptive graph only	3.319	2.362	0.884
METR-LA	Independent masking only	3.496	2.508	0.820
METR-LA	No outage augmentation	3.705	2.657	0.740
METR-LA	No reliability conditioning	3.256	2.339	0.906
METR-LA	No regime router	3.372	2.392	0.872
METR-LA	Static graph only	3.469	2.448	0.841
PEMS-BAY	ShiftGuard	1.948	1.130	0.978
PEMS-BAY	Adaptive graph only	1.884	1.108	0.983
PEMS-BAY	Independent masking only	2.168	1.204	0.935
PEMS-BAY	No outage augmentation	2.651	1.735	0.917
PEMS-BAY	No reliability conditioning	1.881	1.098	0.979
PEMS-BAY	No regime router	1.927	1.115	0.977
PEMS-BAY	Static graph only	2.056	1.238	0.952

Table 8: Local GPU efficiency; latency uses batch one and throughput/memory use batch eight. FLOPs are profiler-observed operations per sample.

Dataset	Model	Parameters	FLOPs/sample	Latency B1 (ms)	Throughput B8	Memory B8 (MiB)
METR-LA	DCRNN	63372	845712576.00	7.97	1052.04	38.27
METR-LA	Graph WaveNet	22744	189736407.00	1.86	7136.17	47.52
METR-LA	STID	41996	4040706.00	0.47	16822.62	34.97
METR-LA	STAEformer	1258932	6080479272.00	2.62	502.59	180.93
METR-LA	ShiftGuard	89012	38541287.00	2.30	3567.45	36.95
PEMS-BAY	DCRNN	63372	1806417600.00	7.47	1045.04	42.00
PEMS-BAY	Graph WaveNet	25104	387135775.00	1.50	6071.88	57.62
PEMS-BAY	STID	49548	6344066.00	0.49	16769.73	36.60
PEMS-BAY	STAEformer	1372212	10386051000.00	3.68	268.54	319.69
PEMS-BAY	ShiftGuard	100340	76656703.00	2.43	3335.28	42.18

7. Discussion

The results isolate a narrower and more defensible novelty than the initial architecture suggested. Structured outage augmentation, calibrated scales, and forecast-origin risk control jointly provide a large operational advantage; reliability-weighted propagation and multi-expert routing do not. This negative ablation result prevents attributing gains to architectural complexity and suggests a lean successor. The formal binomial certificate assumes i.i.d. origin-level failures within each calibration window; under temporal drift we interpret observed coverage and failure prequentially, not as a distribution-free future guarantee. The benchmarks are old, highway-centric, and rely on heuristic sensor graphs, so conclusions are limited to the tested setting.

8. Conclusion

ShiftGuard reframes traffic forecasting under sensor failure as a joint point, interval, and selective-decision problem. It does not improve clean accuracy universally, but across 640 stress evaluations it is materially more robust than locally reproduced baselines and retains substantially more forecasts under a registered risk constraint. The evidence identifies structured outage training and prequential origin-level calibration—not added routing complexity—as the reproducible contribution. Evaluation on newer urban networks and genuinely prospective failures is the necessary next validity test.

Data and code availability

The accompanying repository contains preprocessing, configurations, seeds, checkpoints, predictions, stress masks, and table-generation scripts. Raw datasets are referenced by DOI and excluded from source control.

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